EXISTENCE FOR THE VALUATIVE CRITERION IMPLIES THAT THE MORPHISM IS UNIVERSALLY CLOSED

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- The base change reduction for the category of dotted arrows. Invoked at the start of the counted proof (source lines 9794-9797). Allows the diagram to be replaced by one over a suitable base change.
- The description of the category of dotted arrows and of its set of isomorphism classes. Used throughout the counted proof (source lines 9798-9820). Turns the existence hypothesis into the existence of a dotted arrow.
- The statement that a morphism satisfying existence in the valuative criterion is universally closed. Invoked near the end of the counted proof (source lines 9821-9832). Supplies the conclusion once the hypothesis has been reformulated.

Source reference. Original definition definition-fill-in-diagram: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 9061–9061. Navigation pointer only: the counted text refers to the Stacks item definition-fill-in-diagram, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-cat-dotted-arrows-base-change: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 9187–9187. Navigation pointer only: the counted text refers to the Stacks item lemma-cat-dotted-arrows-base-change, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

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Source reference. Original lemma lemma-quasi-compact-permanence: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 2044–2044. Navigation pointer only: the counted text refers to the Stacks item lemma-quasi-compact-permanence, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-reach-points-scheme-theoretic-image: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 2101–2101. Navigation pointer only: the counted text refers to the Stacks item lemma-reach-points-scheme-theoretic-image, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Lemma 0.1. *Let $f : \mathcal{X} \to \mathcal{Y}$ be a morphism of algebraic stacks. Assume*

(1) *f is quasi-separated, and*
(2) *f is universally closed.*

Then f satisfies the existence part of the valuative criterion.

Proof. Consider a solid diagram

$$\begin{array}{ccc} \mathrm{Spec}(K) & \xrightarrow{x} & \mathcal{X} \\ {\scriptstyle j}\downarrow & \nearrow & \downarrow{\scriptstyle f} \\ \mathrm{Spec}(A) & \xrightarrow{y} & \mathcal{Y} \end{array}$$

where A is a valuation ring with field of fractions K and $\gamma : y \circ j \longrightarrow f \circ x$ as in Definition definition-fill-in-diagram. By Lemma lemma-cat-dotted-arrows-base-change in order to find a dotted arrow (after possibly replacing K by an extension and A by a valuation ring dominating it) we may replace $\mathcal{Y}$ by $\mathrm{Spec}(A)$ and $\mathcal{X}$ by $\mathrm{Spec}(A) \times_{\mathcal{Y}} \mathcal{X}$. Of course we use here that being quasi-separated and universally closed are preserved under base change. Thus we reduce to the case discussed in the next paragraph.

Consider a solid diagram

$$\begin{array}{ccc} \mathrm{Spec}(K) & \xrightarrow{x} & \mathcal{X} \\ {\scriptstyle j}\downarrow & \nearrow & \downarrow{\scriptstyle f} \\ \mathrm{Spec}(A) & = & \mathrm{Spec}(A) \end{array}$$

where A is a valuation ring with field of fractions K as in Definition definition-fill-in-diagram. By Lemma lemma-quasi-compact-permanence and the fact that f is quasi-separated we have that the morphism x is quasi-compact. Since f is universally closed, we have in particular that $|f|(\overline{\{x\}})$ is closed in $\mathrm{Spec}(A)$. Since this image contains the generic point of $\mathrm{Spec}(A)$ there exists a point $x' \in |\mathcal{X}|$ in the closure of x mapping to the closed point of $\mathrm{Spec}(A)$. By Lemma lemma-reach-points-scheme-theoretic-image we can find a commutative diagram

$$\begin{array}{ccc} \mathrm{Spec}(K') & \longrightarrow & \mathrm{Spec}(K) \\ \downarrow & & \downarrow \\ \mathrm{Spec}(A') & \longrightarrow & \mathcal{X} \end{array}$$

such that the closed point of $\mathrm{Spec}(A')$ maps to $x' \in |\mathcal{X}|$. It follows that $\mathrm{Spec}(A') \to \mathrm{Spec}(A)$ maps the closed point to the closed point, i.e., A' dominates A and this finishes the proof. $\square$